1

```
def vacuum_agent(location, status):

    if status == "Dirty":
        return "Suck"
    elif location == "A":
        return "Move Right"
    else:
        return "Move Left"

rooms = {"A": "Dirty", "B": "Clean"}
location = "A"

for i in range(4):
    print("Location:", location, "| Status:", rooms[location])

    action = vacuum_agent(location, rooms[location])
    print("Action:", action)

    if action == "Suck":
        rooms[location] = "Clean"
    elif action == "Move Right":
        location = "B"
    elif action == "Move Left":
        location = "A"

    print()
```

```
# A* Search for 8 Puzzle

start = [1, 2, 3,
         4, 0, 6,
         7, 5, 8]

goal = [1, 2, 3,
        4, 5, 6,
        7, 8, 0]


def heuristic(state):
    count = 0
    for i in range(9):
        if state[i] != 0 and state[i] != goal[i]:
            count += 1
    return count

def get_neighbors(state):
    neighbors = []
    pos = state.index(0)

    moves = {
        0: [1, 3], 1: [0, 2, 4], 2: [1, 5],
        3: [0, 4, 6], 4: [1, 3, 5, 7], 5: [2, 4, 8],
        6: [3, 7], 7: [4, 6, 8], 8: [5, 7]
    }
```

```
    for m in moves[pos]:
        new_state = state[:]
        new_state[pos], new_state[m] = new_state[m], new_state[pos]
        neighbors.append(new_state)

    return neighbors

open_list = [(heuristic(start), start)]
visited = []

while open_list:
    open_list.sort()
    cost, current = open_list.pop(0)

    print(current)

    if current == goal:
        print("\nGoal Reached")
        break

    visited.append(current)

    for neighbor in get_neighbors(current):
        if neighbor not in visited:
            h = heuristic(neighbor)
            open_list.append((h, neighbor))
```